
MODEL SELECTION AND VALIDATION REPORT (remove_4_biomass model)

OBSERVATIONS:1716

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_4_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.32	0.48
remove_4_kde_fric	-0.04	0.03
remove_4_kde_feve	-0.19	0.03
sbt_mean	0.20	0.10
sbt_sd	-0.02	0.05
sst_sd	0.03	0.04
log_chla_mean	0.03	0.03
depth	0.03	0.05
log_sum_fishing	0.09	0.03

Dispersion parameter: 0.85

Spatiotemporal AR1 correlation (rho): 0.97

Matérn range: 383.55

Spatiotemporal marginal AR1 SD: 1.25

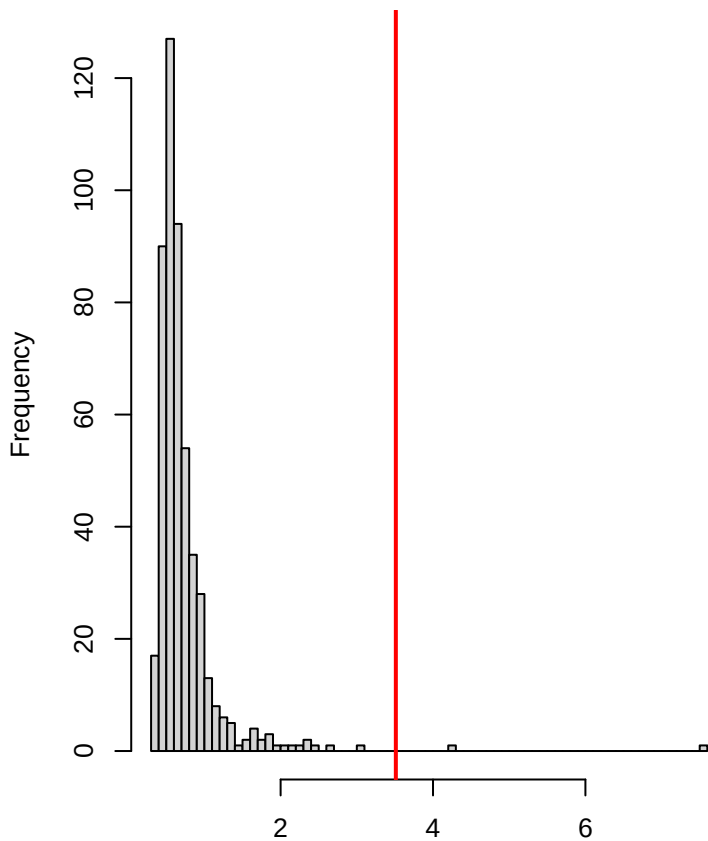
REML criterion at convergence: 12227.246

See ?tidy.sdmTMB to extract these values as a data frame.

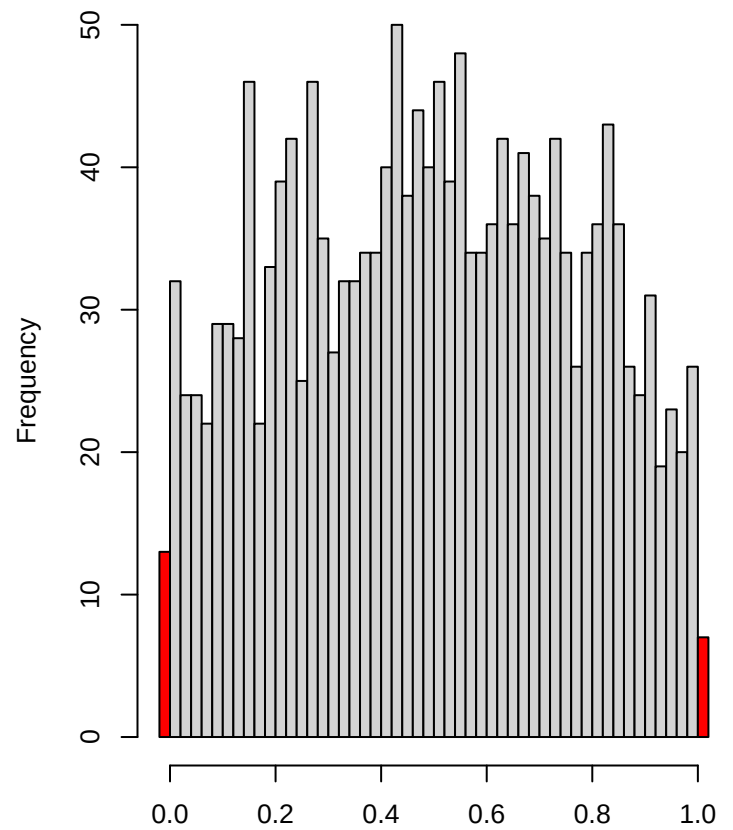
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	24480.49	3.461975
spatiotemporal_random_fields_ar1_with_anisotropy	15	24483.95	135.296293
spatial_random_fields	12	24619.25	2.946551
spatial_random_fields_with_anisotropy	14	24622.20	116.871187
spatiotemporal_random_fields_iid_with_anisotropy	14	24739.07	1.430542
spatiotemporal_random_fields_iid	12	24740.50	246.018607
error_only	10	24986.52	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated

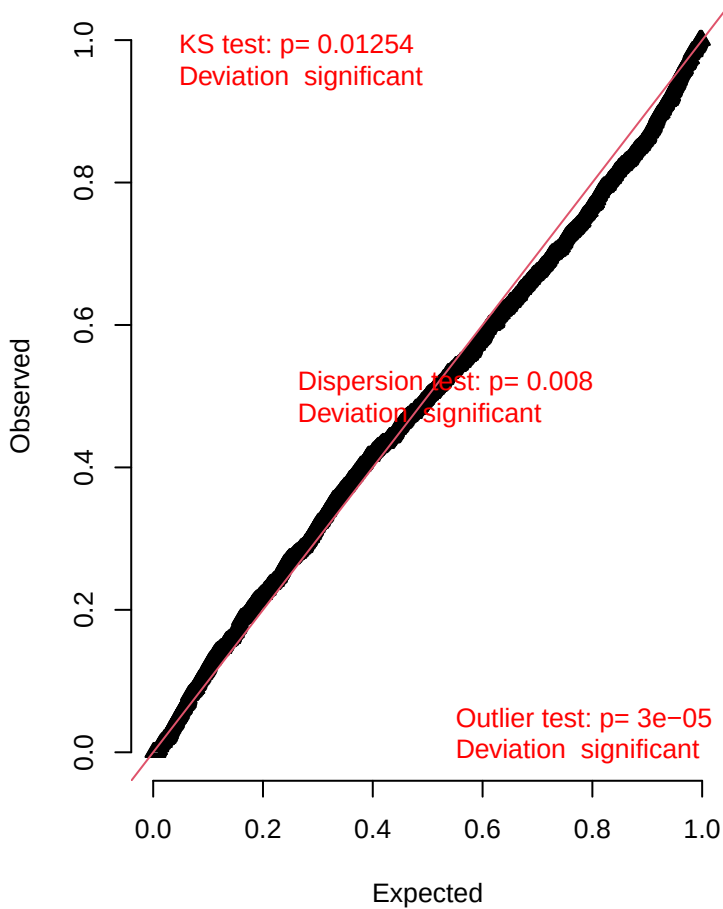


Outlier test significant



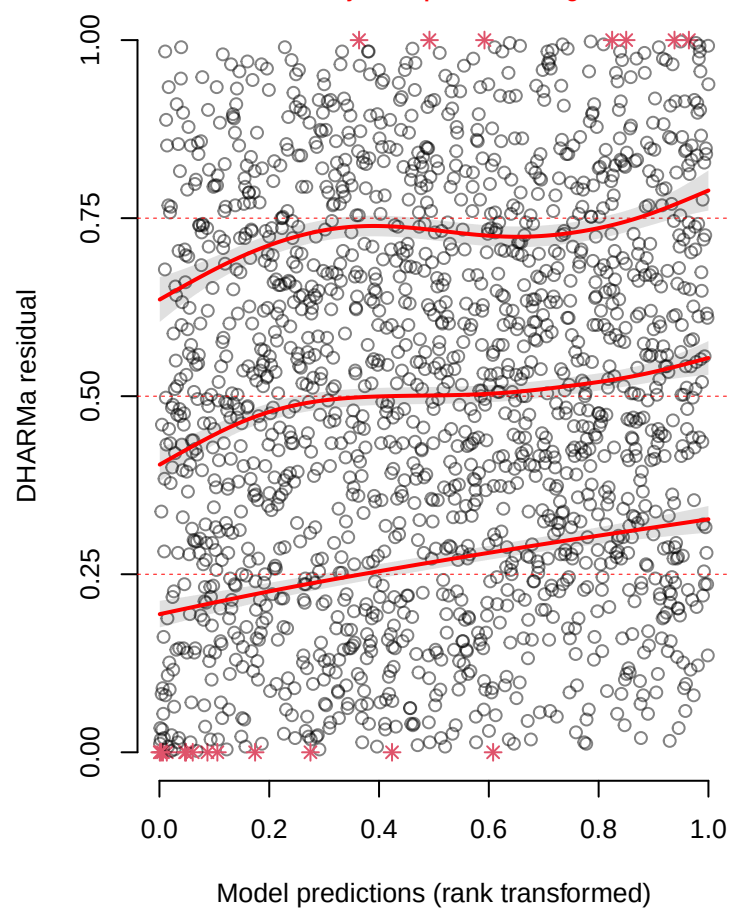
Simulated values, red line = fitted model. p-value (two.sided) =

QQ plot residuals

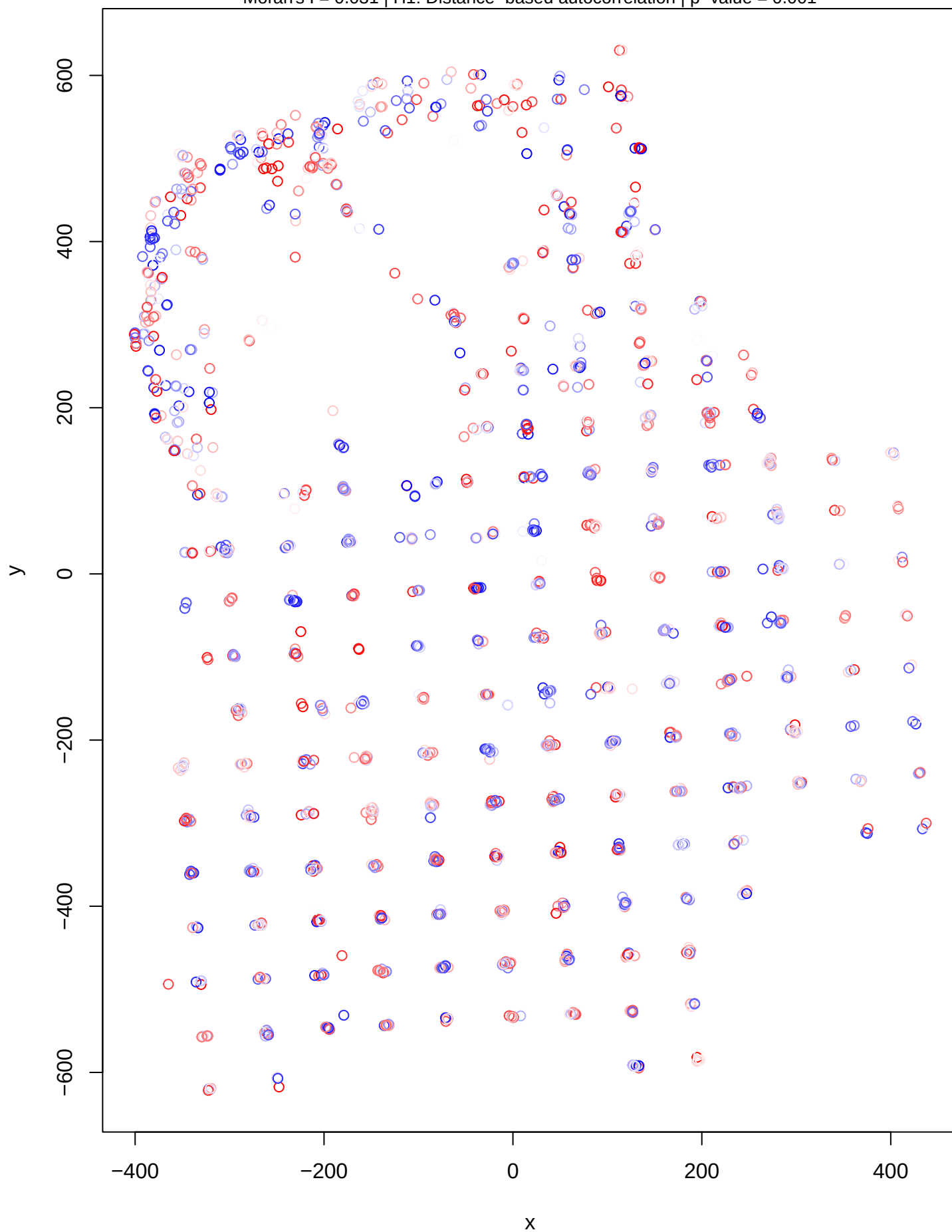


Residuals (outliers are marked red)

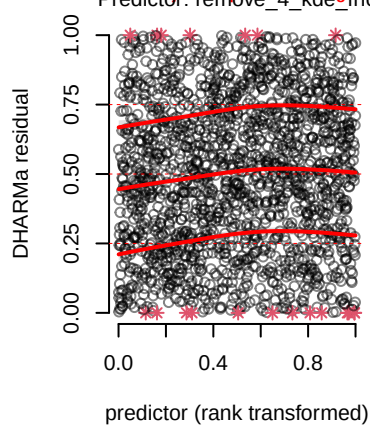
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



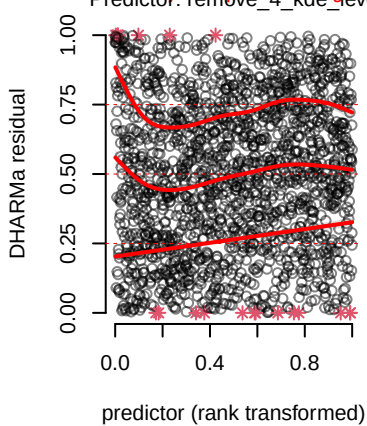
DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.031 | H1: Distance-based autocorrelation | p-value = 0.001



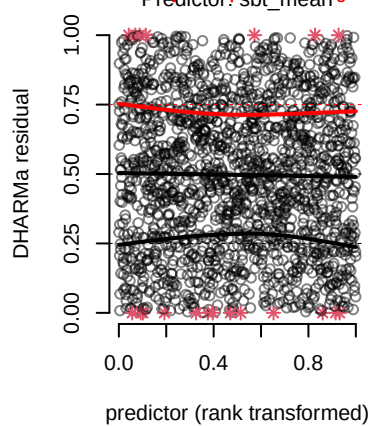
DHARMA residual Residual vs. predictor
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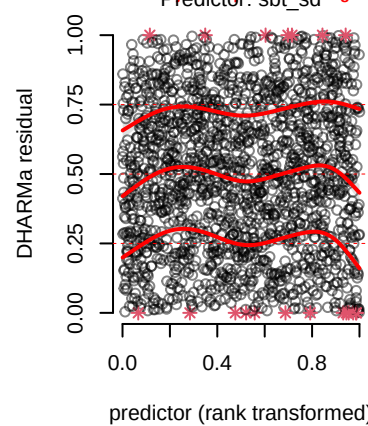
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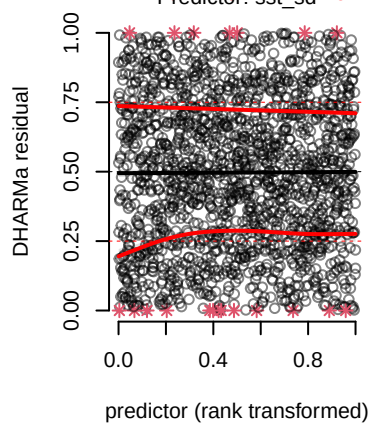
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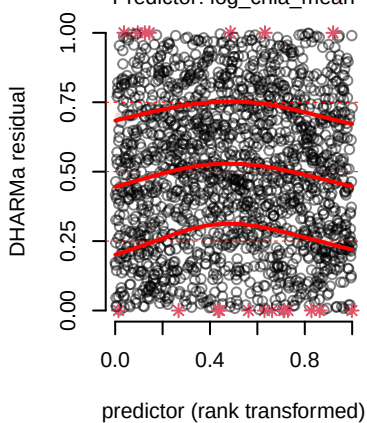
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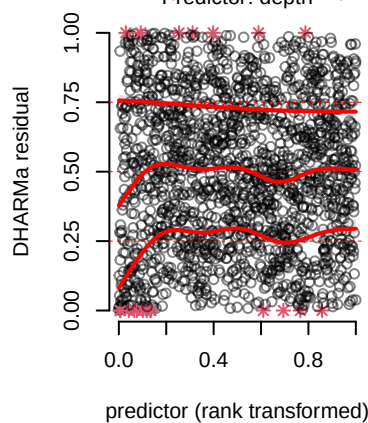
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